

ALL LIFE SPEAKS

WANWU LAIXIN

ENVIRONMENTAL + MULTISPECIES AI

A governance belt that listens,
verifies and replies.

ALS STACK 1.0 / SPATIAL X COMPUTATIONAL TWIN

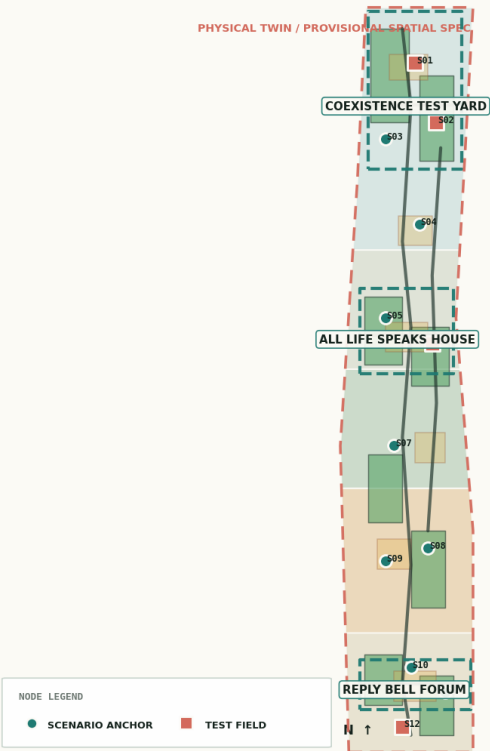
CONCEPT SPEC / NOT DEPLOYED / BENCHMARK REQUIRED

01 · SPATIAL x COMPUTATIONAL TWIN

ONE CITY, TWO SYNCHRONIZED READINGS

A provisional spatial plan is paired with a bounded, auditable computational twin architecture.

CONCEPT SPEC · NOT DEPLOYED · TWIN NOT CALIBRATED



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01



PROVISIONAL CONCEPT · NOT A STATUTORY PLAN

A CIVIC LETTER IS AN AUDITABLE OBJECT

THESIS

AI does not impersonate species. It compiles environmental evidence into a public object with provenance, uncertainty, a named human owner and a reversible status.

PERMISSION BOUNDARY

Candidate detection, anomaly finding, evidence retrieval and drafting are allowed. Planning approval, engineering findings, health diagnosis, policing, enforcement, case closure and critical-system control are forbidden.

CONCEPT SPEC

NOT DEPLOYED

BENCHMARK REQUIRED

HUMAN RELEASE

ROLLBACK / EXIT

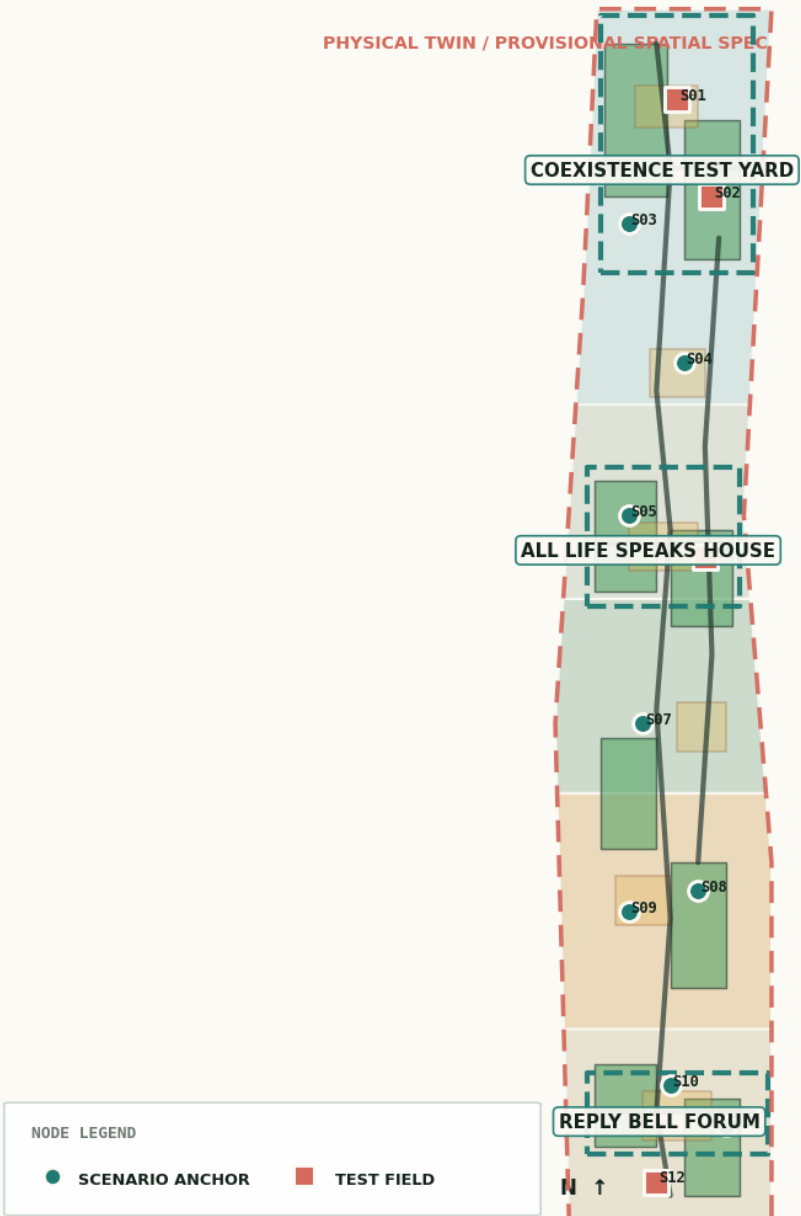
LIFE LETTER SCHEMA / v0.1 CONCEPT

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{
  "letter_id": "LETTER-XXX",
  "signal_domain": "environmental",
  "space_feature_id": "SCENE-XXX",
  "model_family": "candidate_not_selected",
  "model_version": null,
  "confidence_status": "benchmark_required",
  "ood_flag": "unknown",
  "source_provenance": "required",
  "human_reviewer_role": "named_domain_role",
  "fallback_mode": "manual_or_fixed_rule",
  "decision_status": "verification_required"
}
```

ONE CITY, TWO SYNCHRONIZED READINGS

A provisional spatial plan is paired with a bounded, auditable computational twin architecture.

CONCEPT SPEC · NOT DEPLOYED · TWIN NOT CALIBRATED



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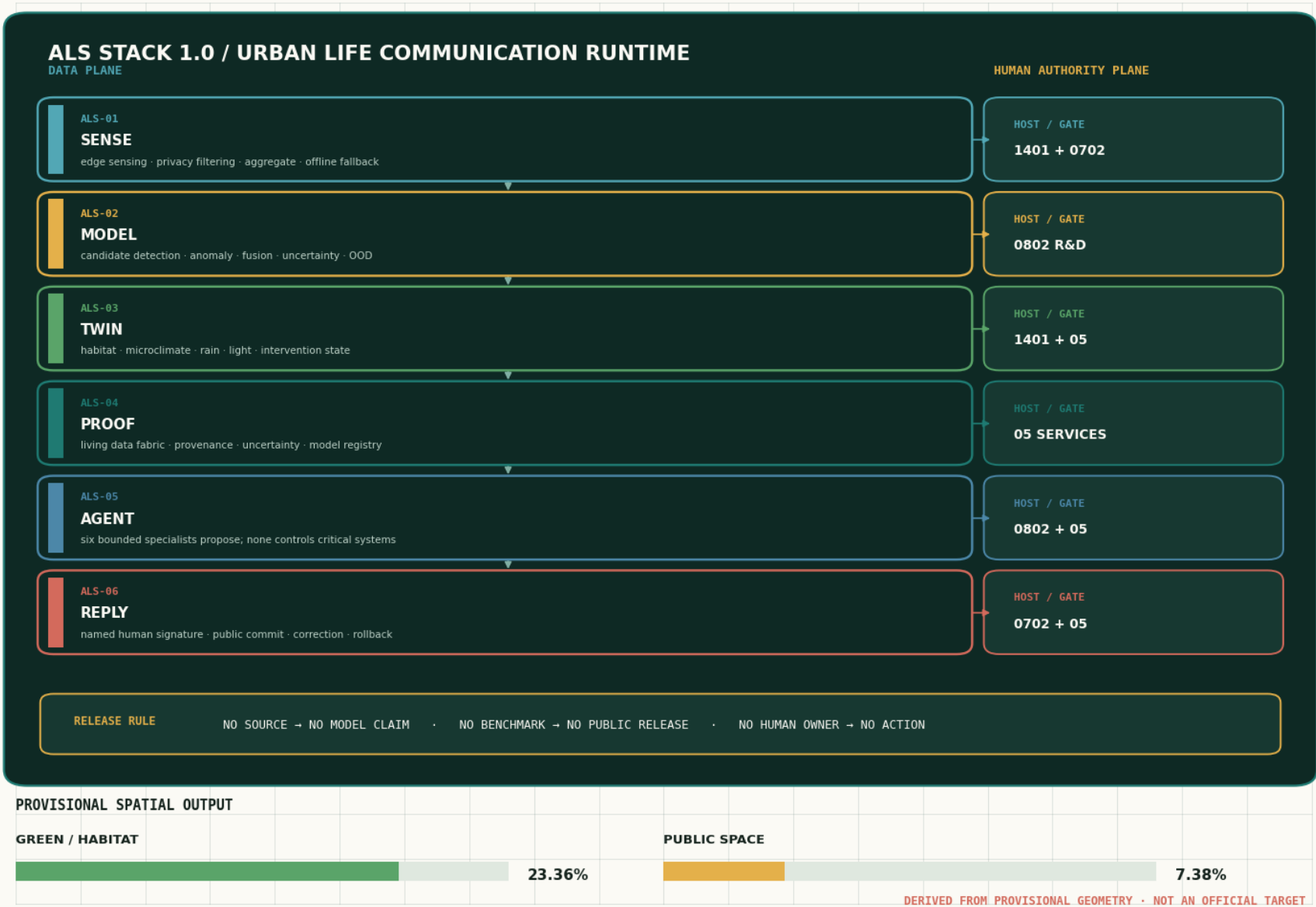
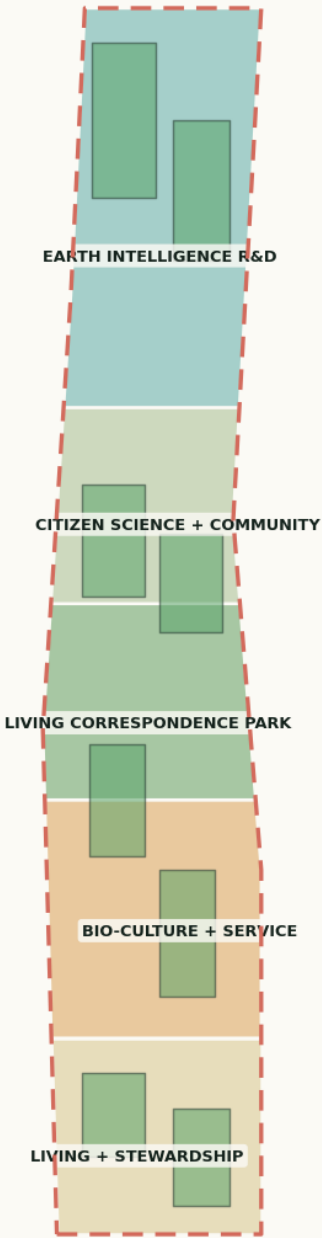


PROVISIONAL CONCEPT · NOT A STATUTORY PLAN

FIVE URBAN BANDS × SIX COMPUTE LAYERS

Land, public space and bounded AI are designed as one open interface specification.

ARCHITECTURE SPEC · MODEL NOT SELECTED · BENCHMARK REQUIRED



Three provisional key areas host different stages of one bounded urban-AI verification system.

LAB PROGRAMME SPEC · REVERSIBLE COMPONENTS · PROFESSIONAL REVIEW REQUIRED

01

ZHONGZHIYUAN

LAB

COEXISTENCE TEST YARD

PROVISIONAL
KEY AREA

NOT A DEVICE PLAN

LAB PIPELINE / PHYSICAL → COMPUTE → CIVIC

01

HARDWARE

low-power multimodal bench

02

COMPUTE

edge privacy scrubber

03

TEST

drift · outage · robot e-stop

04

PUBLIC

model cards + failure archive

HUMAN GATE

STOP / raw identifiable data leaves device or safe shutdown fails

02

AI ORIGIN COMMUNITY

LAB

ALL LIFE SPEAKS HOUSE

PROVISIONAL
KEY AREA

NOT A DEVICE PLAN

LAB PIPELINE / PHYSICAL → COMPUTE → CIVIC

01

INTERFACE

Life Letter compiler

02

KNOWLEDGE

expert-reviewed graph

03

CORRECT

citizen feedback → test set

04

ACCESS

paper + staff + digital

HUMAN GATE

RELEASE / expert confirms evidence; non-digital service stays available

03

DAZHONGSI

LAB

REPLY BELL FORUM

PROVISIONAL
KEY AREA

NOT A DEVICE PLAN

LAB PIPELINE / PHYSICAL → COMPUTE → CIVIC

01

TWIN

façade + light scenarios

02

LOG

versioned civic commit wall

03

REVIEW

accept · revise · reject

04

ROLLBACK

pause · restore · exit

HUMAN GATE

CULTURAL GATE / concept only; heritage, safety and ecology review required

ALL LIFE SPEAKS · JINGZHANG

PROVISIONAL CONCEPT · NOT A STATUTORY PLAN

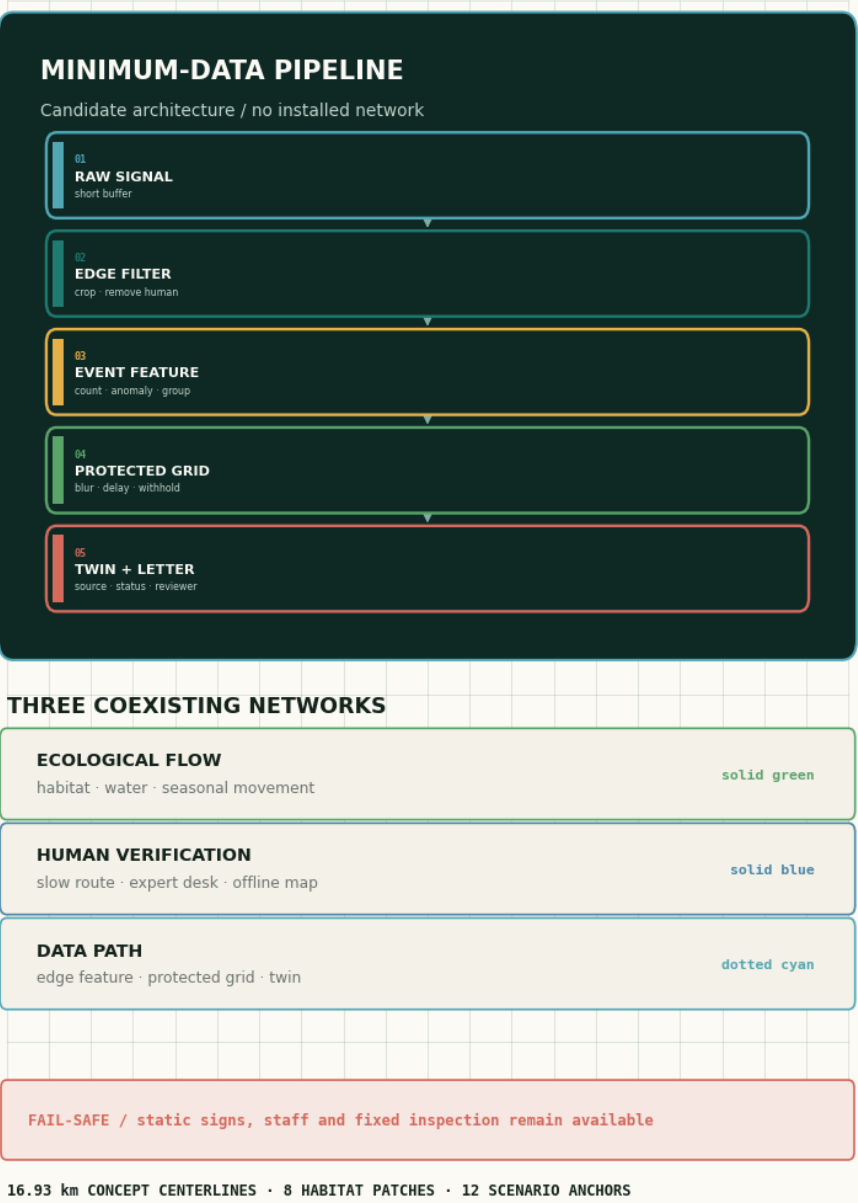
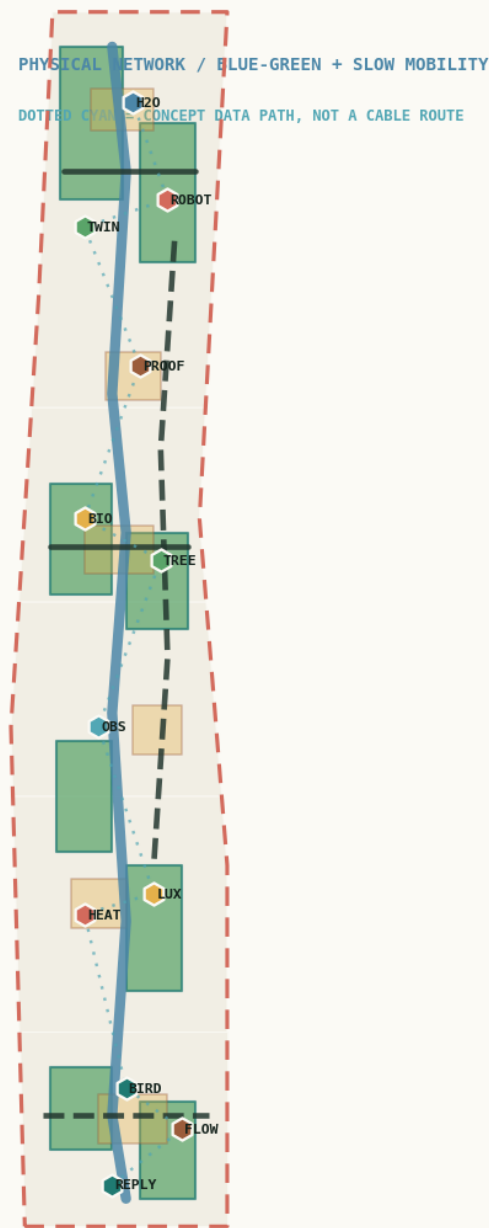
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PROVISIONAL CONCEPT · NOT A STATUTORY PLAN

THREE NETWORKS, ONE MINIMUM-DATA PATH

Ecological flow, human verification and candidate data links share a reversible public-space system.

SCENARIO ANCHORS · NOT DEVICE LOCATIONS · NO WIRELESS COVERAGE CLAIM



KNOWN GEOMETRY, UNKNOWN PERFORMANCE

A release dashboard separates reproducible design counts from unmeasured model and service performance.

NO TRAINED MODEL · NO APPROVED TEST SET · NO MEASURED AI PERFORMANCE

SITE AREA

11.41 km²

EPSG:4548 · provisional

GREEN / HABITAT

23.36 %

8 concept patches

PUBLIC SPACE

7.38 %

6 concept polygons

SLOW NETWORK

16.93 km

5 concept centerlines

SCENE NODES

12

4 industry validations

PERSONAS

6

digital + non-digital

PLANNING UNKNOWN / OFFICIAL POLYGONS · FAR · HEIGHT · BIODIVERSITY BASELINE · HEAT EFFECT · STORMWATER CAPACITY

AI BENCHMARK LEDGER / ALL VALUES UNMEASURED

CALIBRATION ERROR

UNKNOWN ECE

approved field test set absent

FALSE ALERT RATE

UNKNOWN FAR

ground-truth operations absent

OOD RECALL

UNKNOWN OOD-R

shift corpus not approved

EDGE P95 LATENCY

UNKNOWN P95

hardware + model unselected

ENERGY / INFERENCE

UNKNOWN J

device profile unmeasured

HUMAN REVIEW COVERAGE

UNKNOWN %

no operational log

FALLBACK AVAILABILITY

UNKNOWN %

no service operation

APPEAL RESOLUTION

UNKNOWN h

no public-case baseline

FAILURE-INJECTION SUITE / SPECIFIED, NOT YET EXECUTED

T-01 SENSOR LOSS

T-02 CLOCK DRIFT

T-03 SEASON / OOD

T-04 HUMAN MEDIA CAPTURE

T-05 SENSITIVE LOCATION

T-06 CAUSAL HALLUCINATION

T-07 NETWORK / POWER LOSS

T-08 REVIEWER DISAGREEMENT

EVIDENCE RELEASE CHAIN



RULE / UNKNOWN STAYS UNKNOWN UNTIL DATA, PROTOCOL, BENCHMARK AND NAMED HUMAN REVIEW EXIST.

TWELVE MACHINE-READABLE LETTERS

01 WATER

SIGNAL water time series / MODEL anomaly candidate /
HUMAN water expert / FALLBACK sampling

02 RAIN

SIGNAL rain + asset state / MODEL forecast candidate /
HUMAN operator / FALLBACK inspection

03 TREE

SIGNAL canopy + soil / MODEL vision-time candidate /
HUMAN arborist / FALLBACK ledger

04 SOIL

SIGNAL chain-of-custody sample / MODEL clustering aid /
HUMAN lab / FALLBACK test

05 POLLINATOR

SIGNAL bounded view / MODEL group candidate / HUMAN
entomologist / FALLBACK transect

06 BIRD

SIGNAL de-voiced audio / MODEL acoustic candidate /
HUMAN expert / FALLBACK observation

07 NIGHT

SIGNAL lux + safety / MODEL comparison candidate /
HUMAN safety-lighting / FALLBACK rules

08 SMALL FAUNA

SIGNAL low-res thermal / MODEL event candidate / HUMAN
ecology-traffic / FALLBACK field

09 HEAT

SIGNAL microclimate / MODEL exposure hint / HUMAN
operator / FALLBACK static map

10 FOOD

SIGNAL aggregate flow / MODEL forecast candidate /
HUMAN community / FALLBACK weighing

11 CITIZEN REPLY

SIGNAL voluntary media / MODEL multimodal candidate /
HUMAN expert + citizen / FALLBACK paper

12 GOVERNANCE

SIGNAL verified proof / MODEL source-bound draft / HUMAN
owner / FALLBACK manual case

SIX BOUNDED ASSISTANTS, ONE HUMAN AUTHORITY PLANE

AG-01 WATER-AGENT

Organizes water and microclimate anomalies and field-review queues. No compliance or health conclusion.

AG-02 HABITAT-AGENT

Combines candidate habitat signals. Never exposes protected locations or claims species increase.

AG-03 NBS-OPS-AGENT

Drafts inspection options. Cannot close cases, penalize staff or control critical assets.

AG-04 EVIDENCE-GUARD

Checks provenance, model version, OOD and failure status. Cannot confer professional approval.

AG-05 CIVIC-RAG-AGENT

Retrieves registered public sources, preserves minority views and never treats a draft as law.

AG-06 REPLY-ORCHESTRATOR

Surfaces conflict and drafts a review packet. Cannot approve, enforce, fund or close a case.

HUMAN AUTHORITY PLANE

Ecology · planning · operations · accessibility · public governance. Release, revise, roll back or exit.

FAILURE INJECTION BEFORE PUBLIC RELEASE

T-01 SENSOR LOSS

Remove one modality. The system must expose missing evidence and fall back to manual observation.

T-02 CLOCK DRIFT

Inject stale or misaligned time. The letter must stop or display an invalid-time state.

T-03 SEASON / OOD

Present distribution shift. The model must abstain or route to expert review.

T-04 HUMAN MEDIA

Inject speech or faces. Raw identifiable content must be removed or the test stopped.

T-05 SENSITIVE LOCATION

Attempt exact-location release. The system must blur, delay or withhold it.

T-06 CAUSAL HALLUCINATION

Draft an unsupported ecological cause. Evidence agent and human gate must reject it.

T-07 POWER / NETWORK

Disconnect edge and cloud. Fixed rules, signage and manual service must remain.

T-08 REVIEW DISAGREEMENT

Submit conflicting expert views. Publish uncertainty and an appeal path, not a false consensus.

TWELVE RELEASE GATES FOR PUBLIC ENVIRONMENTAL AI

01 MINIMIZE

Environmental and non-human signals by default; coarse space; no account when unnecessary.

02 LIMIT PURPOSE

No advertising, profiling, attendance, policing or punitive reuse.

03 PROTECT LOCATION

Nest, breeding and rare-species locations are grid-shifted, delayed or withheld.

04 TRANSLATE HUMBLY

Every output says: provisional AI interpretation of environmental evidence.

05 HUMAN REVIEW

Safety, access, lighting, trees, traffic and operations require domain sign-off.

06 EXPLAIN

Publish source class, confidence, gaps, human edits and decision role.

07 APPEAL

Residents, workers and experts can correct identity, priority, access and impact errors.

08 EXIT

People can delete voluntary content; ethics review can stop a device or trial.

09 FAIL SAFELY

Stop AI output and restore fixed rules, manual inspection and service.

10 KEEP NON-AI

Paper, phone, staff, static signs and manual transects remain available.

11 NO AUTO-PENALTY

No person, merchant, employee, community or operator is automatically ranked or punished.

12 CHECK FAIRNESS

Annual review compares territory, time and accessible-user service coverage.

BASELINE · SANDBOX · PILOT · EVALUATE · SCALE OR EXIT

P0 / BASELINE

Obtain authorized geometry, field protocols, licences, ecological baselines and named human owners.

P1 / SANDBOX

Run synthetic and controlled failure injection. No public release and no critical-system control.

P2 / REVERSIBLE PILOT

Only reviewed scenarios enter a small public test with non-AI fallback, appeal and stop controls.

P3 / EVALUATE

Publish known, unknown, model cards, failures, human edits and territorial service coverage.

HARD BOUNDARY

Official polygons, current buildings, rights, roads, utilities, heritage controls, FAR, heights, engineering and ecological baselines are missing.

DECISION

Named humans recommend scale, revise, pause or exit. Every outcome becomes a versioned public commit log.